Food Traceability and Genomics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Traceability and Genomics (A07841)

Short Title:	Food Traceability and Genomics		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	SWALSH		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module introduces the student to the concerns arising from the increasing globalisation of the food supply chain, and addresses the systems used to ensure food safety and traceability throughout the supply chain, from producer to consumer. In addition, the students will learn to apply their knowledge of genetics and genomics to food traceability. The module will highlight the regulations within the EU and will focus on current and emerging methods of food traceability, both electronic and diagnostic (assays and molecular).

Programmes

SCIE-0064	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	4 / 7 / M
SCIE-0064	BSc (Hons) in Food Science and Innovation (WD_SFDSC_B)	4 / 1 / M

Indicative Content

- Emerging food safety, traceability and security issues
- Traceability systems
- DNA-based diagnostic methodologies such as PCR, gel electrophoresis etc.
- Non-molecular based methodologies for food authentication
- Immunoassay technologies to identify adulteration
- Review their own learning progress using continuous assessment methodologies
- Comparative analyses of relevant literature articles

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate food traceability, safety and security issues and their influence on future food production.
2. Appraise different molecular technologies and their application in food/feed safety and traceability systems.
3. Compare and contrast various non-molecular methods of analyses for use in food authentication.
4. Evaluate the principles of immunoanalysis and its relevance to food safety and traceability.
5. Appraise the legislative systems governing food/feed traceability (RASFF, TRACES etc.).
6. Assess various laboratory techniques to determine food/feed authentication.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Guest industry speakers.
- Case studies will be employed to reinforce module material.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Lab Report	40%	6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.
40%-49%: Will be able to show a good knowledge of the key learning outcomes including a good understanding of factors affecting food traceability and the application of various technologies to food traceability.
50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes as well as being able to describe and evaluate the effectiveness of systems used to ensure food safety, traceability and security.
60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics including legislative systems governing food/feed traceability and demonstrate an ability to relate detailed knowledge acquired to new material.
70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	24	
Independent Learning	87	

Essential Material

- "Food Safety Authority of Ireland." www.fsai.ie
- Bertheau, Y. and J. Davison. _Genetically Modified and non-Genetically Modified Food Supply Chains : Co-Existence and Traceability_. UK: John Wiley & Sons, 2012.
- Bray, D., K. Hopkin, B. Albers, A. Johnson, M. Raff and Roberts, K., Walter, P. _Essential Cell Biology_. USA: Garland Science, 2012.
- Shan, G. _Immunoassays in Agricultural Biotechnology_. USA: John Wiley & Sons, 2010.
- Toldra, F. and L.M.L. Nollet. _Advances in Food Diagnostics_. USA: Wiley-Blackwell, 2007.

Requested Resources

- Room Type: Biology Lab